10/23/2026: Real-valued functions of real variables

CSCI 246: Discrete Structures

Textbook reference: Chap 11.1, Epp

Today's Agenda

- Quiz ($\approx$ 30 mins)
- Group exercises ($\approx$ 20 mins)

Quiz 7

Answer any two questions below for full credit. Answer a third question for extra credit.

1. Prove by smallest counterexample that $n < 2^n$ for all $n \in \mathbb{N}$.
2. Show by induction that $f_0 + \cdots + f_n = f_{n+2} - 1$ in the Fibonacci sequence.
3. Let a be the sequence $0, 2, 7, 15, 26, 40, 57, \ldots$. Write a formula expressing a_n (the n-th term in the sequence) as a polynomial function of n.

Definition. The **Fibonacci sequence** is the sequence given by the following recursive rules:

$$f_0 = 0, \qquad f_1 = 1, \qquad f_{n+2} = f_n + f_{n+1}$$

Theorem 23.17 Let $a_0, a_1, a_2, \ldots$ be a sequence of numbers. The terms a_n can be expressed as polynomial expressions in n if and only if there is a nonnegative integer k such that for all $n \geq 0$ we have $\Delta^{k+1} a_n = 0$. In this case,

$$a_n = a_0 \binom{n}{0} + (\Delta a_0) \binom{n}{1} + (\Delta^2 a_0) \binom{n}{2} + \cdots + (\Delta^k a_0) \binom{n}{k}.$$

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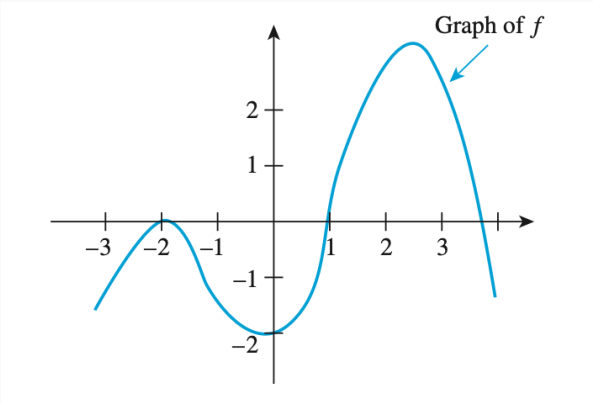
- Lets us handle continuous variables (like runtime).
- Big-O notation is typically defined with real numbers so that tools from calculus can be used (like derivatives and limits). However, we won't use calculus tools in this course.

Random group assignments

Ashley, Isabelle: 4
Bailey, Tanna: 6
Blaisdell, Olyver: 12
Brown, Ashton: 9
Cardenas, Dominic: 5
Cutler, George: 11
Fabik, Roland: 6
Fellows, Audrey: 10
Fitzgerald, Ryan: 4
Fuller, Ryan: 7
Griffen, Clara: 9
Grutsch, Tanner: 1
Hager, Nathan: 11
Horner, Jayden: 12
Johnson, Jaret: 6
Kintzel, Samantha: 7
Lameres, Alexis: 1
Lee, Holden: 4
Mayala, Reece: 3
McLean, Connor: 8
Mcbride, Daniel: 8
Miller, Maverick: 10
Moss, Emily: 2
Mousseau, Matthew: 11
Neuman, Evan: 12
Putnam, John: 2
Ramos, Vydel: 3
Rathbun, Jedidiah: 2
Roberts, Matthew: 10
Roller, Nathan: 1
Ryan, Otis: 9
Shevtsov, Timothy: 8
Stensland, Emma: 5
Thiessen, Logan: 3
Thompson, Jaiden: 5
Vath, Ethan: 7

Group exercises

1. Sketch the graphs of the power functions p_2 and p_3 on the same set of axes.
   a. When $x > 1$, which is greater, x^2 or x^3?
   b. When $0 < x < 1$, which is greater, x^2 or x^3?
2. Repeat the above problem for $p_{1/2}$ and $p_{1/3}$.
3. The graph of the function f is shown below. Find the intervals on which f is increasing and the intervals on which f is decreasing.



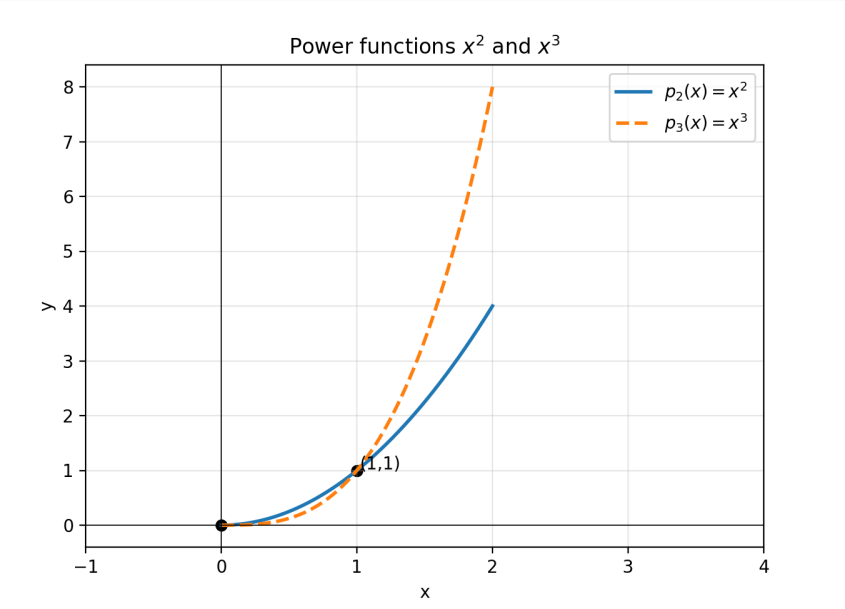
4. Show that if a function $f : \mathbb{R} \to \mathbb{R}$ is increasing, then f is one-to-one.
5. Let f be a real-valued function of a real variable. Show that if f is decreasing on a set S and if M is any positive real number, then Mf is decreasing on S.
6. Let h be a function from $\mathbb{R}$ to $\mathbb{R}$ defined by the formula $h(x) = x^2$ for each real number x.
   a. Show that h is decreasing on the set of real numbers less than zero.
   b. Show that h is increasing on the set of real numbers greater than zero.

Solution to group exercise #1

Problem. Sketch the graphs of the power functions p_2 and p_3 on the same set of axes.

a. When $x > 1$, which is greater, x^2 or x^3?
b. When $0 < x < 1$, which is greater, x^2 or x^3?

Solution.



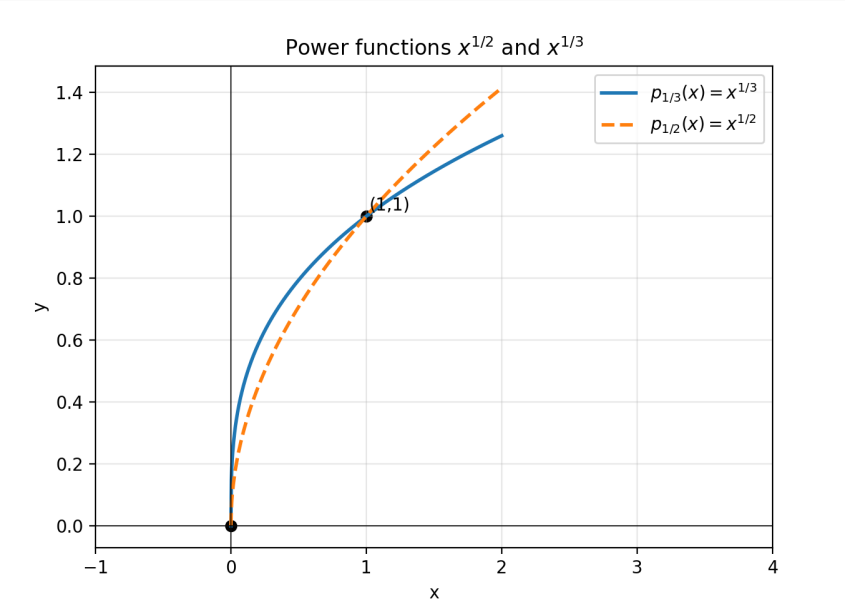
a. x^3. This happens because multiplying by x (which is greater than 1) repeatedly makes the result larger each time. b. x^2. This happens because multiplying by x (which is less than 1) repeatedly makes the result smaller each time.

Solution to group exercise #2

Problem. Sketch the graphs of the power functions $p_{1/2}$ and $p_{1/3}$ on the same set of axes.

a. When $x > 1$, which is greater, $x^{1/2}$ or $x^{1/3}$?
b. When $0 < x < 1$, which is greater, $x^{1/2}$ or $x^{1/3}$?

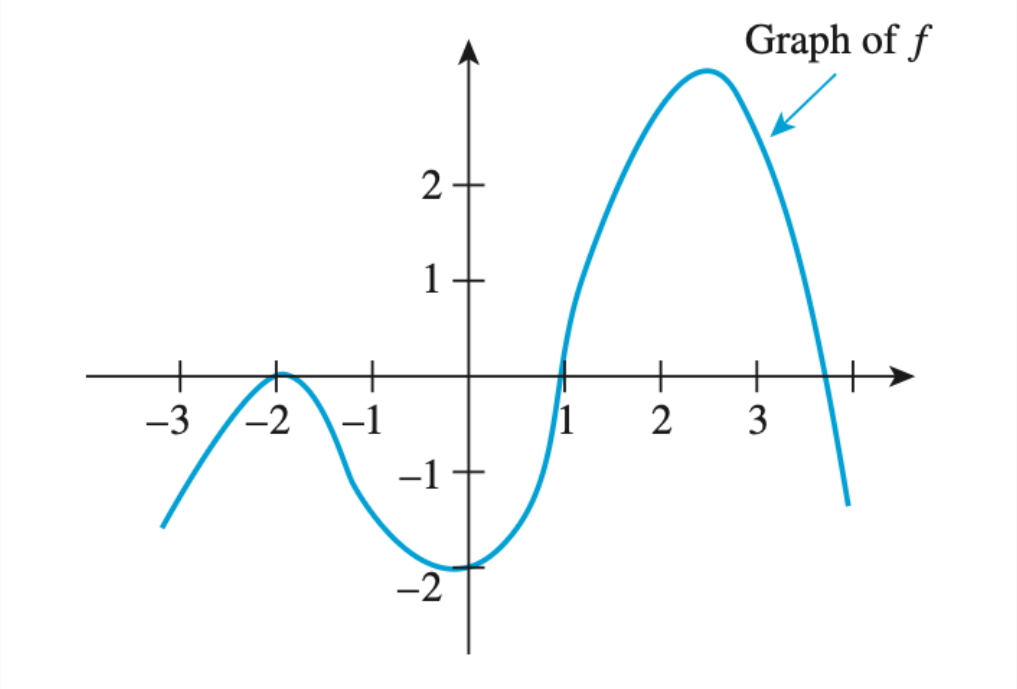
Solution.



a. $x^{1/2}$. b. $x^{1/3}$. *Why?* Write $x^{1/2} = x^{1/3} \cdot x^{1/6}$. If $x > 1$, then $x^{1/6} > 1$, so multiplying by it makes $x^{1/3}$ larger. If $0 < x < 1$, then $0 < x^{1/6} < 1$, so multiplying by it makes $x^{1/3}$ smaller.

Solution to group exercise #3

Problem. The graph of the function f is shown below. Find the intervals on which f is increasing and the intervals on which f is decreasing.



Solution. The function f is

- increasing on $(-3, -2)$ and $(0, 2)$, and
- decreasing on $(-2, 0)$ and $(2, 4)$.

Solution to group exercise #4

Problem. Show that if a function $f : \mathbb{R} \to \mathbb{R}$ is increasing, then f is one-to-one.

Solution. We want to show that f is one-to-one. That is, we want to show that that distinct inputs lead to distinct outputs: if $x \neq y$, then $f(x) \neq f(y)$. Now since x and y are real numbers, if $x \neq y$, then either $x < y$ or $y > x$. So since f is increasing, either $f(x) < f(y)$ or $f(y) > f(x)$. Either way, $f(x) \neq f(y)$.

Solution to group exercise #5

Problem. Let f be a real-valued function of a real variable. Show that if f is decreasing on a set S and if M is any positive real number, then Mf is decreasing on S.

Solution. We have that for all $x, y \in S$,

$$x < y \implies f(x) > f(y) \implies Mf(x) > Mf(y),$$

where the first implication holds since f is decreasing on S, and the second implication holds since multiplication by a positive number preserves the direction of an inequality.

Hence, Mf is decreasing.

Solution to group exercise #6

Problem. Let h be a function from $\mathbb{R}$ to $\mathbb{R}$ defined by the formula $h(x) = x^2$ for each real number x.

a. Show that h is decreasing on the set of real numbers less than zero.

b. Show that h is increasing on the set of real numbers greater than zero.

Solution.

a. Let $x, y < 0$ with $x < y$. Multiplying $x < y$ by x and by y (both negative, so each reverses the inequality) gives

$$x^2 > xy \qquad \text{and} \qquad xy > y^2.$$

Chaining these, $x^2 > xy > y^2$, so $h(x) > h(y)$. Hence h is decreasing on the negative real numbers.

b. Let $x, y > 0$ with $x < y$. Multiplying $x < y$ by x and by y (both positive, so each preserves the inequality) gives

$$x^2 < xy \qquad \text{and} \qquad xy < y^2.$$

Chaining these, $x^2 < xy < y^2$, so $h(x) < h(y)$. Hence h is increasing on the positive real numbers.